Example 3.1

$$X_t = Z_t + \beta_1 Z_{t-1}$$
$$= (1 + \beta_1 B) Z_t \qquad (B = \text{backward shift operator})$$
$$= \theta(B) Z_t$$

where $\theta(B) = 1 + \beta_1 B$.

The process is invertible if the roots of $\theta(z) = 0$ lie outside the unit circle.

Let $1 + \beta_1 z = 0$

$$\Rightarrow z = -\frac{1}{\beta_1}$$

$-\frac{1}{\beta_1}$ lies outside the unit circle where

$$\left| -\frac{1}{\beta_1} \right| > 1 \quad \Rightarrow \quad |\beta_1| < 1.$$

Hence the process $\{X_t\}$ is invertible if $|\beta_1| < 1$.